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Modeling tidal stream energy extraction and its effects on transport processes in a tidal channel and bay system using a three-dimensional coastal ocean model

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Abstract

This paper presents a numerical modeling study for simulating in-stream tidal energy extraction and assessing its effects on the hydrodynamics and transport processes in a tidal channel and bay system connecting to coastal ocean. A marine and hydrokinetic (MHK) module was implemented in a three-dimensional (3-D) coastal ocean model using the momentum sink approach. The MHK model was validated with the analytical solutions for tidal channels under one-dimensional (1-D) conditions. Model simulations were further carried out to compare the momentum sink approach with the quadratic bottom friction approach. The effects of 3-D simulations on the vertical velocity profile, maximum extractable energy, and volume flux reduction across the channel were investigated through a series of numerical experiments. 3-D model results indicate that the volume flux reduction at the maximum extractable power predicted by the 1-D analytical model or two-dimensional (2-D) depth-averaged numerical model may be overestimated. Maximum extractable energy strongly depends on the turbine hub height in the water column, which reaches a maximum when turbine hub height is located at mid-water depth. Far-field effects of tidal turbines on the flushing time of the tidal bay were also investigated. Model results demonstrate that tidal energy extraction has a greater effect on the flushing time than volume flux reduction, which could negatively affect the biogeochemical processes in estuarine and coastal waters that support primary productivity and higher forms of marine life.

Highlights

► Model validation of tidal energy extraction against an analytical solution. ► 3-D effects on velocity, power extraction and volume flux. ► Volume flux reduction predicted by 1-D and 2-D models may be overestimated. ► Effects of tidal energy extraction on flushing time.

Introduction

In response to growing concerns over global climate change, there has been increasing interest in extraction of tidal stream energy from coastal waters, using tidal currents as a carbon-free renewable energy source. One advantage of tidal energy is its high predictability in comparison to other renewable energy sources, such as terrestrial winds and solar energy. Traditional tidal energy extraction is based on a tidal barrage approach in which tidal power is generated by the head difference during ebb and flood tides. Due to many environmental concerns with the tidal barrage approach, extracting tidal energy through in-stream tidal turbines has gained more attention in theoretical research and technology development in recent years [1], [2], [3], [4], [5].

An increasing number of analytical and numerical modeling studies have been conducted recently to evaluate the amount of extractable power from a tidal system without considering the potential environmental impacts on the tidal system. Garrett and Cummins [6] developed a one-dimensional (1-D) analytical model to investigate the maximum extractable tidal energy in a tidal channel as a function of tidal amplitude and volume flux across the channel cross-section, assuming that turbines fully occupy the entire cross-section of the channel. Garrett and Cummins [7] further examined the extractable tidal power with turbines partially occupying the cross-section. Using a similar approach to Garrett and Cummins [6], Blanchfield et al. [8] presented an analytical model for estimating the extractable power in a channel linked to a bay. Expanding on Garrett and Cummins' work [6], [7], Vennell [9] demonstrated that optimal tunings of turbines in a tidal channel depend on the turbine array configurations, channel geometry, background bottom friction, and tidal forcing. Atwater and Lawrence [10] and Polagye and Malte [11] subsequently extended a similar 1-D modeling approach to tidal systems with a split tidal channel and more complex channel networks, respectively.

There are also a few studies on tidal power extraction using two-dimensional (2-D) numerical models that incorporate the tidal turbine in the model to assess tidal energy extraction. Sutherland et al. [12] applied a depth-averaged 2-D model to assess tidal energy extraction in Johnstone Strait off Vancouver Island. Sun et al. [13] simulated the local effects on the flow field due to tidal energy extraction at a tidal site in Puget Sound using the computational fluid dynamics code FLUENT. Karsten et al. [14] applied the Finite Volume Coastal Ocean Model (FVCOM) in 2-D mode to simulate the tidal current energy in the Minas Passage, Bay of Fundy. Walkington and Burrows [15] simulated the tidal stream power potential and impacts of installed tidal farm capacity on the west coast of the United Kingdom. Draper et al. [16] applied a depth-averaged 2-D numerical model to qualify the effects of tidal energy extraction on tidal hydrodynamics in a simple tidal channel.

Because tidal flows are generally three-dimensional (3-D) in nature, a 3-D modeling approach will provide the most realistic assessment of extractable power and the associated effects on water circulation and the transport processes in coastal environments. Most recently, 3-D models have been used to study tidal energy extraction at specific sites. For example, Shapiro [17] used a 3-D ocean circulation model to estimate the maximum extractable energy, change of current magnitudes and effects on passive tracer transport due to energy extraction. Define et al. [18] modeled the effects of tidal power extraction on the estuarine hydrodynamics along the Georgia coast. Hasegawa et al. [19] evaluated the far-field effects of tidal energy extraction in Minas Passage, Bay of Fundy using a 3-D nested-grid tidal circulation model.

Although there are a growing number of numerical modeling studies of tidal energy extraction, studies using 3-D models are still limited; particularly those that examine the 3-D effects on tidal energy extraction and flushing time. This paper describes the implementation and validation of a marine and hydrokinetic (MHK) module in a 3-D unstructured-grid finite volume coastal ocean model. Simulation of tidal energy extraction in the model is directly related to tidal turbine characteristics, including turbine dimensions, thrust coefficient, and drag coefficients associated with supporting structures. The model validation against the analytical solution of tidal energy extraction in tidal channels [6], [8] is presented. Sensitivity modeling analysis to investigate the difference between 2-D and 3-D modeling approaches on maximum extractable energy, velocity profiles, and volume flux were conducted. The effect of tidal energy extraction on flushing time and its implication for changes in transport processes was also simulated and discussed.

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Section snippets

Energy extraction using a momentum sink approach

There are two common approaches to simulate the effect of tidal hydrokinetic energy extraction using analytical and numerical models. The first is the bottom friction approach, in which the tidal energy dissipation associated with the presence of tidal turbines is approximated by increasing bottom friction [6], [10], [12]. This approach is commonly used in analytical models and depth-averaged 2-D numerical model simulations. The second approach is the momentum sink approach, in which a momentum ...

Analytical model of energy extraction in a tidal channel

One of the challenges of modeling tidal energy extraction is model validation because observation data associated with tidal energy extraction are extremely limited. In this paper, the MHK module embedded in FVCOM is validated against the 1-D analytical model for a tidal channel developed by Garrett and Cummins [6] and Blanchfield et al. [8]. Assuming the tidal channel is fully occupied by tidal turbines, Garrett and Cummins [6] found that power P can be related to the volume flux across the ...

Three-dimensional effect on the velocity profile

For the purpose of model validation against the 1-D analytical solution, model runs in Section 3 were conducted in a depth-averaged 2-D mode. However, in reality, tidal turbines are deployed in a water body with 3-D variations in the flow field. To investigate the 3-D effects, model simulations in a 3-D mode were conducted for the same turbine configuration described in Section 3. In the 3-D model setup, twenty-one uniform vertical layers in a sigma-stretched coordinate were specified, from the ...

Discussion and summary

In this study, an MHK module for tidal energy extraction was incorporated into a 3-D unstructured-grid Finite Volume Coastal Ocean Model FVCOM using the momentum sink approach. The momentum sink terms due to energy extraction were directly represented by the turbine thrust coefficient, turbine dimensions, and associated supporting structures. The MHK model was validated in an idealized tidal channel connecting to a bay against the analytical model initially developed by Garrett and Cummins [6] ...

Acknowledgements

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The cumulative impact of tidal stream turbine arrays on sediment transport in the Pentland Firth
2015, Renewable Energy

Citation Excerpt :
...Tidal stream turbines can only be deployed in specific resource areas where tidal current velocities exceed a certain device dependant level, typically these are in areas constrained by channels, islands or headlands [2–5]. Implementation of turbine arrays is likely to lead to areas of reduced tidal current due to energy extraction and also accelerated tidal flows in other areas due to blockage effects [6–8]. If mobile sediment is present, this might lead to changes in sediment transport regime and also to the morphology of sandy areas....

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Hydrokinetic energy conversion systems: A technology status review
2015, Renewable and Sustainable Energy Reviews

Citation Excerpt :
...One dimensional (1-D) analytical models are used for the effects on water level and velocity, whereas, advanced 2-D and 3-D are implemented to calculate the source potential [18]. Yang et al. [17] gives an updated list of models that employed to determine the tidal stream resources.
Majority of developed models are based on tidal power....

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